DAT 412: Machine Learning Final Project

1 Project Overview

For your final project, you will develop a comprehensive machine learning solution to a real-world problem of your choice. This project will integrate various concepts and techniques learned throughout the course, demonstrating your ability to apply machine learning algorithms to solve practical problems.

2 Project Requirements

1. Choose a dataset and define a problem to solve using machine learning techniques.
2. Implement at least three different machine learning algorithms.
3. Perform data preprocessing, feature engineering, and model evaluation.
4. Compare and analyze the performance of different models.
5. Present your findings in a detailed report and prepare a presentation.

3 Step-by-Step Plan

3.1 Step 1: Problem Definition and Dataset Selection (Week 1-2)

- Research and select a dataset from reputable sources (e.g., Kaggle, UCI Machine Learning Repository).
- Define the problem you want to solve and formulate it as a machine learning task.
- Identify the type of learning (supervised, unsupervised, or reinforcement) required.

3.2 Step 2: Data Preprocessing and Exploratory Data Analysis (Week 3-4)

- Load and inspect the dataset using Jupyter Notebook.
- Perform data cleaning, handling missing values, and outlier detection.
- Conduct exploratory data analysis using visualization techniques (Matplotlib, Seaborn).
- Apply feature engineering and selection techniques.

3.3 Step 3: Model Selection and Implementation (Week 5-7)

- Choose at least three appropriate machine learning algorithms for your problem.
- Implement the selected algorithms using Python and relevant libraries (e.g., scikit-learn, TensorFlow).
- For supervised learning tasks, split the data into training and testing sets.
- Train the models on the preprocessed data.

3.4 Step 4: Model Evaluation and Comparison (Week 8-9)

- Evaluate model performance using appropriate metrics (e.g., accuracy, precision, recall, F1-score for classification; MSE, R-squared for regression).
- Implement cross-validation to ensure robust performance estimation.
- Compare the performance of different models using learning curves and other visualization techniques.
- Analyze bias and variance trade-offs for each model.

3.5 Step 5: Model Optimization (Week 10-11)

- Apply regularization techniques to prevent overfitting.
- Perform hyperparameter tuning using techniques like grid search or random search.
- If applicable, implement ensemble methods (e.g., Random Forest, XGBoost) to improve performance.
- Reassess model performance after optimization.

3.6 Step 6: Final Analysis and Report Preparation (Week 12-14)

- Summarize your findings, including the problem statement, methodology, results, and conclusions.
- Discuss the strengths and limitations of your approach.
- Prepare visualizations to effectively communicate your results.
- Write a comprehensive report detailing your entire process and findings.

4 Evaluation Criteria

Your project will be evaluated based on the following criteria:

- Problem formulation and dataset selection (10%)
- Data preprocessing and exploratory data analysis (15%)
- Implementation of machine learning algorithms (25%)
- Model evaluation and comparison (20%)
- Analysis and interpretation of results (15%)
- Report quality (15%)

5 Submission Guidelines

- Submit your final report as a PDF document.
- Include all code files and Jupyter notebooks used in your project.
- Provide a README file with instructions to reproduce your results.
- Due date: May 2, 11.59pm